

Data types & Vectors & For Loops

ds4owd - data science for openwashdata

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Learning Objectives (for this week)

1. Learners can identify different ways of accessing a vector from a dataframe
2. Learners can list the four main atomic vector types in R.
3. Learners can create cross-references to figures and tables using labels and captions in Quarto documents
4. Learners can recognize when a for loop is appropriate for repetitive tasks in data analysis

Part 1: Cross-references

Cross references

- Help readers to navigate your document with numbered references and hyperlinks to entities like figures and tables.
- Cross referencing steps:
 - Add a caption to your figure or table.
 - Give an ID to your figure or table, starting with `fig-` or `tbl`
 - Refer to it with `@fig-...` or `@tbl-....`

Table cross references

The presence of the caption ([A few penguins](#)) and label (`#tbl-penguins`) make this table reference-able:

See `@tbl-penguins` for data on a few penguins.

becomes:

See [Table 1](#) for data on a few penguins.

```
1 ```{r}
2 #| label: tbl-penguins
3 #| tbl-cap: A few penguins
4
5 head(penguins) |>
6   gt()
7 ```
```

Table 1: A few penguins

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007
Adelie	Torgersen	36.7	19.3	193	3450	female	2007

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
Adelie	Torgersen	39.3	20.6	190	3650	male	2007

Figure cross references

The presence of the caption ([A few penguins](#)) and label ([#fig-penguins](#)) make this figure reference-able:

See [@fig-penguins](#) for data on a few penguins.

becomes:

See [Figure 1](#) for data on a few penguins.

```
1 ```{r}
2 #| label: fig-penguins
3 #| fig-cap: A few penguins
4 #| fig-width: 6
5 #| fig-asp: 0.618
6
7 ggplot(penguins, aes(x = species, fill = species)) +
8   geom_bar(show.legend = FALSE)
9 ```
```

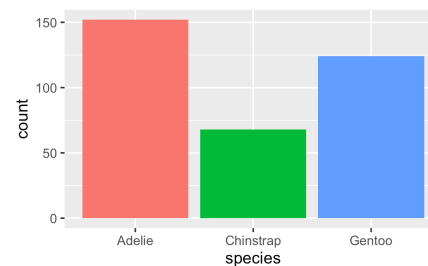


Figure 1: A few penguins

Your turn: md-06-exercises

1. Open posit.cloud/spaces/663318/content in your browser.
2. Verify that the ds4owd workspace is open.
3. Click Start next to md-06-exercises.
4. In the File Manager in the bottom right window, locate the [01a-cref-your-turn.qmd](#) file and click on it to open it in the top left window.

Part 2: Data types and vectors

Why care about data types?

Example: survey data

A survey about recycling behaviour in the city of Zurich:

- `job`: What is your occupation?
- `price_glass`: What monthly subscription would you be willing to pay for a metal/glass bin outside your home?

id	job	price_glass
1	Student	0
2	Retired	0
3	Other	0
4	Employed	10
5	Employed	See comment
6	Student	5-10
7	Student	0
8	Retired	0
9	Student	10
10	Employed	0  ds4owd-002.github.io/website/

id	job	price_glass
11	Employed	20 (2CHF per person with 10 people in the WG)
12	Student	10
13	Student	10
14	Employed	0
15	Student	10
16	Student	0
17	Employed	5-10
18	Other	0
19	Student	0
20	Employed	10
21	Employed	0
22	Employed	5

Oh why won't you work?!

```
1 survey_data_small |>
2   summarise(mean_price_glass = mean(price_glass))

# A tibble: 1 × 1
  mean_price_glass
              <dbl>
1                NA
```

Oh why won't you still work??!!

```
1 survey_data_small |>
2   summarise(mean_price_glass = mean(price_glass, na.rm = TRUE))

# A tibble: 1 × 1
  mean_price_glass
      <dbl>
1             NA
```

Take a breath and look at your data

id	job	price_glass
1	Student	0
2	Retired	0
3	Other	0
4	Employed	10
5	Employed	See comment
6	Student	5-10
7	Student	0
8	Retired	0
9	Student	10
10	Employed	0
11	Employed	20 (2CHF per person with 10 people in the WG)
12	Student	10
13	Student	10
14	Employed	0 🔗 ds4owd-002.github.io/website/

id	job	price_glass
15	Student	10
16	Student	0
17	Employed	5-10
18	Other	0
19	Student	0
20	Employed	10
21	Employed	0
22	Employed	5

Take a breath and look at your data

```
# A tibble: 22 × 3
   id job      price_class
  <int> <chr>    <chr>
1     1 Student  0
2     2 Retired  0
3     3 Other    0
4     4 Employed 10
5     5 Employed See comment
6     6 Student  5-10
7     7 Student  0
8     8 Retired  0
9     9 Student  10
10    10 Employed 0
# i 12 more rows
```

Very common data tidying step!

```
1 survey_data_small |>
2   mutate(price_glass_new = case_when(
3     price_glass == "5-10" ~ "7.5",
4     price_glass == "05-Oct" ~ "7.5",
5     str_detect(price_glass, pattern = "2CHF") == TRUE ~ "20",
6     str_detect(price_glass, pattern = "See comment") == TRUE ~ NA_character_,
7     TRUE ~ price_glass
8   ))
```

Very common data tidying step!

id	job	price_glass_new	price_glass
1	Student	0	0
2	Retired	0	0
3	Other	0	0
4	Employed	10	10
5	Employed	NA	See comment
6	Student	7.5	5-10
7	Student	0	0
8	Retired	0	0
9	Student	10	10
10	Employed	0	0
11	Employed	20	20 (2chf per person with 10 people in the WG)
12	Student	10	10
13	Student	10	10

id	job	price_glass_new	price_glass
14	Employed	0	0
15	Student	10	10
16	Student	0	0
17	Employed	7.5	5-10
18	Other	0	0
19	Student	0	0
20	Employed	10	10
21	Employed	0	0
22	Employed	5	5

Sumamrise? Argh!!!!

```
1 survey_data_small |>
2   mutate(price_glass_new = case_when(
3     price_glass == "5-10" ~ "7.5",
4     price_glass == "05-Oct" ~ "7.5",
5     str_detect(price_glass, pattern = "20") == TRUE ~ "20",
6     str_detect(price_glass, pattern = "See comment") == TRUE ~ NA_character_,
7     TRUE ~ price_glass
8   )) |>
9   summarise(mean_price_glass = mean(price_glass_new, na.rm = TRUE))
```

```
# A tibble: 1 × 1
  mean_price_glass
      <dbl>
1             NA
```

Always respect your data types!

Taking the mean of vector with type “character” is not possible.

```
# A tibble: 22 × 4
```

	id	job	price_glass	price_glass_new
	<int>	<chr>	<chr>	<chr>
1	1	Student	0	0
2	2	Retired	0	0
3	3	Other	0	0
4	4	Employed	10	10
5	5	Employed	See comment	<NA>
6	6	Student	5-10	7.5
7	7	Student	0	0
8	8	Retired	0	0
9	9	Student	10	10
10	10	Employed	10	10

Always respect your data types!

```
1 survey_data_small |>
2   mutate(price_glass_new = case_when(
3     price_glass == "5-10" ~ "7.5",
4     price_glass == "05-Oct" ~ "7.5",
5     str_detect(price_glass, pattern = "20") == TRUE ~ "20",
6     str_detect(price_glass, pattern = "See comment") == TRUE ~ NA_character_,
7     TRUE ~ price_glass
8   )) |>
9   mutate(price_glass_new = as.numeric(price_glass_new)) |>
10  summarise(mean_price_glass = mean(price_glass_new, na.rm = TRUE))
```

```
# A tibble: 1 × 1
  mean_price_glass
      <dbl>
1           4.76
```


My turn: Types of vectors & For loops

Sit back and enjoy!

Take a break

Please get up and move! Let your emails rest in peace.



Your turn: md-06-exercises

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Take a break

Please get up and move! Let your emails rest in peace.



Homework assignments module 6

Module 6 documentation

[ds4owd-002.github.io/website/content/modules/
md-06.html](https://ds4owd-002.github.io/website/content/modules/md-06.html)

Module 6

Data types & Vectors & For Loops

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🎯 Learning Objectives

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🖥 Slides

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Homework due date

- Homework assignment due: November 05, 2025

Wrap-up

Thanks!

Slides created via revealjs and Quarto: <https://quarto.org/docs/presentations/revealjs/>

Access slides as [PDF on GitHub](#)

References

Ben Aleya, Ali, Daniel Biek, Lin Boynton, Julia Jaeggi, Sebastian Camilo Loos, Chiara Mey Piening, Jonathan Olal Ogwang, et al. 2022. “Research Beyond the Lab, Spring Term 2022, Global Health Engineering, ETH Zurich. Raw Data and Analysis-Ready Derived Data on Waste Management in Public Spaces in Zurich, Switzerland.” Zenodo. <https://doi.org/10.5281/ZENODO.7331120>.